

```
Text",10)).pack()

Yentry = Label(w, text = "Y Co-ordinate = ").pack()
yv = StringVar()
yentry = Entry(w, textvariable = yv, font=("Sitka
Text",10)).pack()

MAGentry = Label(w, text = "Magnitude of Charge = ")
).pack()
magv = StringVar()

magentry = Entry(w, textvariable = magv, font=("Sitka
Text",10)).pack()

load = Button(w, text = 'ADD CHARGE DETAILS', command =
Appender).pack(pady=10)

def Plotter():

    # plot field lines
    x0, x1=-10, 10
    y0, y1=-10, 10
    x=linspace(x0, x1, 64)
    y=linspace(y0, y1, 64)
    x, y=meshgrid(x, y)
    Ex, Ey=E_total(x, y, charges)
    streamplot(x, y, Ex, Ey, color='k')
```

```
# plot point charges
for C in charges:
    if C.q>0:
        plot(C.pos[0], C.pos[1], 'bo', ms=8*sqrt(C.q))
    if C.q<0:
        plot(C.pos[0], C.pos[1], 'ro', ms=8*sqrt(-C.q))

    xlabel('$x$')
#labelling x axis
    ylabel('$y$')
# labelling y axis
    gca().set_xlim(x0, x1)
    gca().set_ylim(y0, y1)
    show()
```

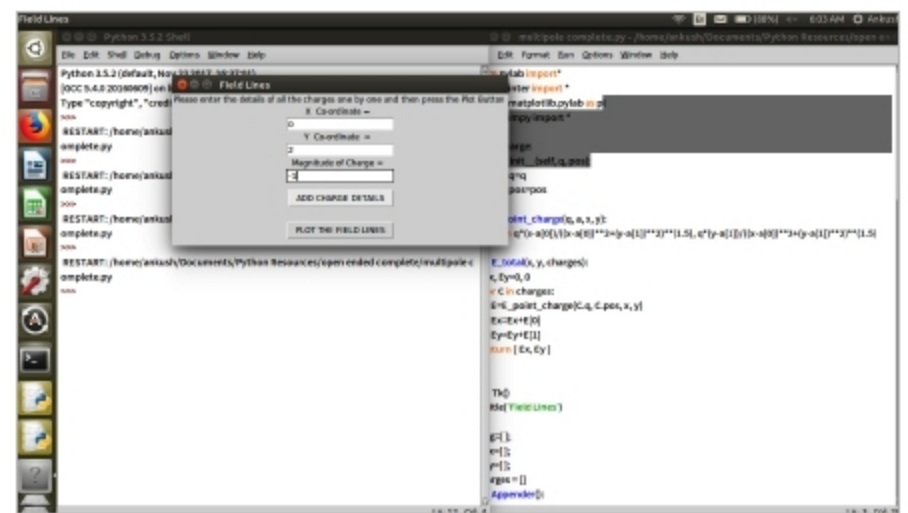


Figure 4: Adding details of the third charge

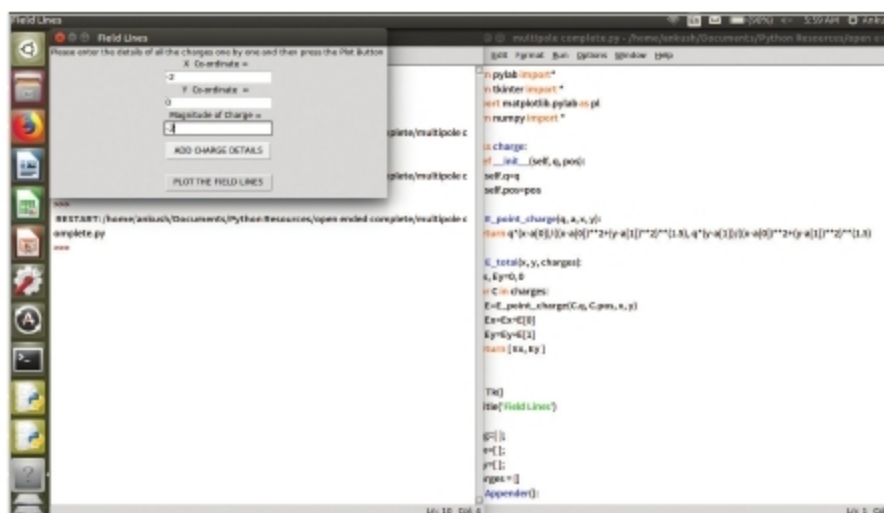


Figure 2: Adding details of the first charge

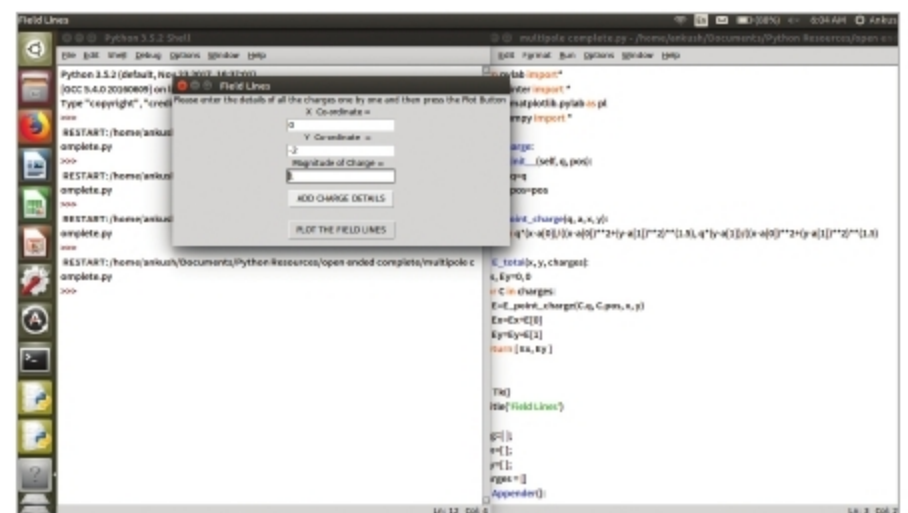


Figure 5: Adding details of the fourth charge

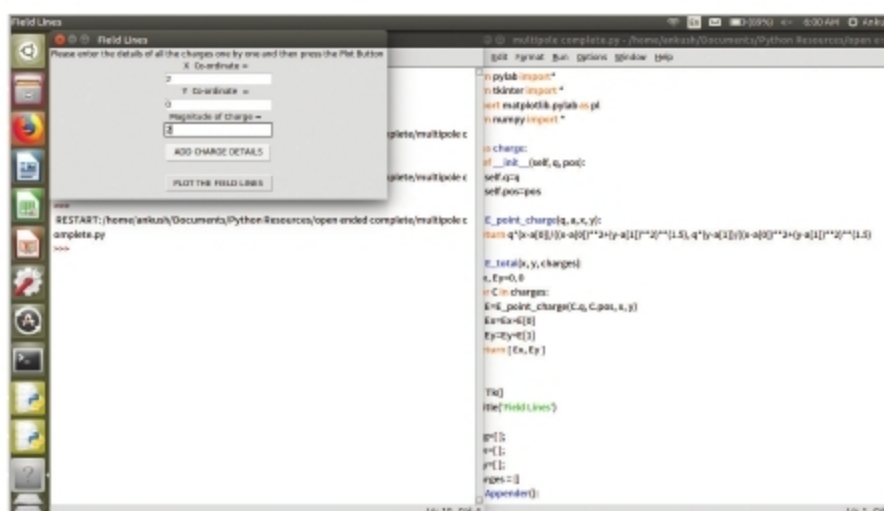
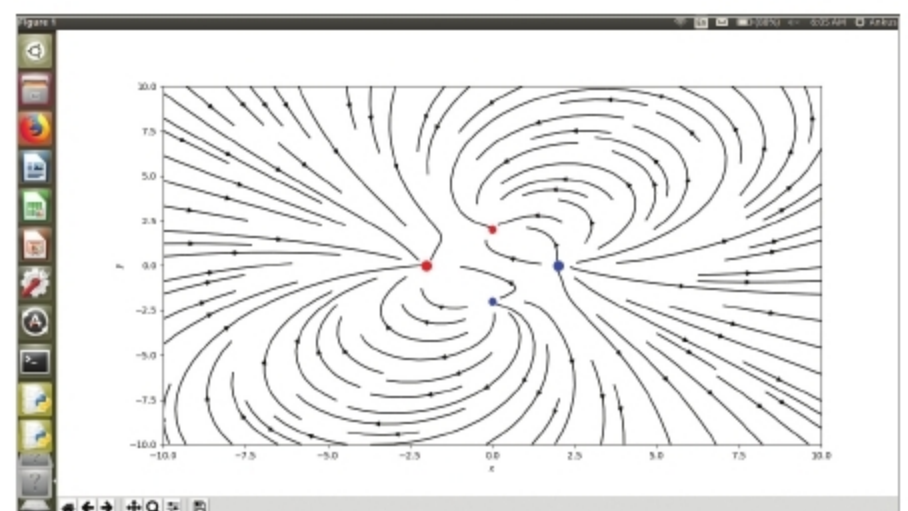


Figure 3: Adding details of the second charge

Figure 6: Output obtained on pressing the *plot field lines* button